

Unit 12, Lesson 3: Writing Algebraic Expressions

Benchmark

MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.

Learning Target

- ☐ I can translate a written description into an algebraic expression.

12.3.1 Warm-Up: Bell Work

1. Two diagrams are shown. One represents $3 + 5 = 8$. The other represents $5 \cdot 3 = 15$.

Complete the table by writing the equation that matches each diagram in the space provided and labeling the length of each diagram.

Equation		
Diagram	33333	35

2. Draw a diagram that represents each equation.

a. $4 + 3 = 7$

b. $4 \cdot 3 = 12$

12.3.2 Refresher: Sums, Differences, Products, and Quotients

1. Complete each statement.

The

- ☐ difference
- ☐ product
- ☐ quotient
- ☐ sum

 of 4 and 6 is 10 because $4 + 6 = 10$.

The

- ☐ difference
- ☐ product
- ☐ quotient
- ☐ sum

 of 8 and 3 is 5 because $8 - 3 = 5$.

The

- ☐ difference
- ☐ product
- ☐ quotient
- ☐ sum

 of 32 and 2 is 16 because $\frac{32}{2} = 16$.

The

- ☐ difference
- ☐ product
- ☐ quotient
- ☐ sum

 of 7 and 9 is 63 because $7 \times 9 = 63$.

12.3.3 Guided Instruction: Making Sense with Models

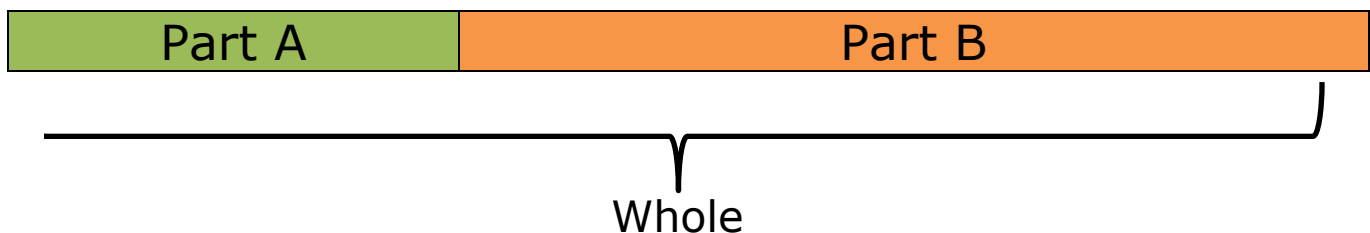
An **algebraic expression** is a mathematical statement containing numbers, operators, and at least one unknown, represented by a variable.

- Complete the statement.

An expression
☐ does
☐ does not
 have an equal sign or inequality symbol.

Bar models help connect the action of a problem to an algebraic expression.

Part-Part-Whole Model



The part-part-whole model can be used to show the following actions.

Action	Equation
Combining or joining	$\text{Part A} + \text{Part B} = \text{Whole}$
Taking away or separating	$\text{Whole} - \text{Part A} = \text{Part B}$ $\text{Whole} - \text{Part B} = \text{Part A}$

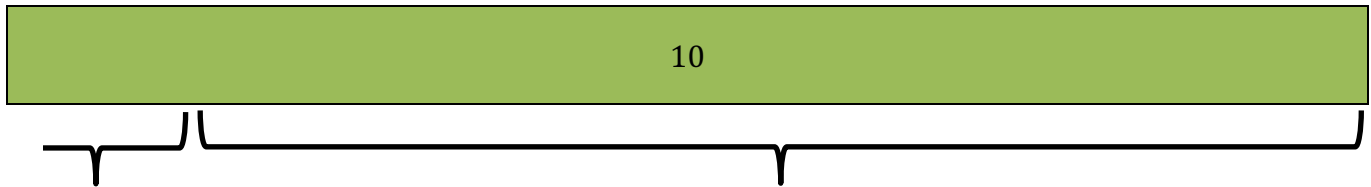
- “The sum of a number and three” can be shown as the length of the whole bar where one part is n and the other part is 3.

The sum of a number and three



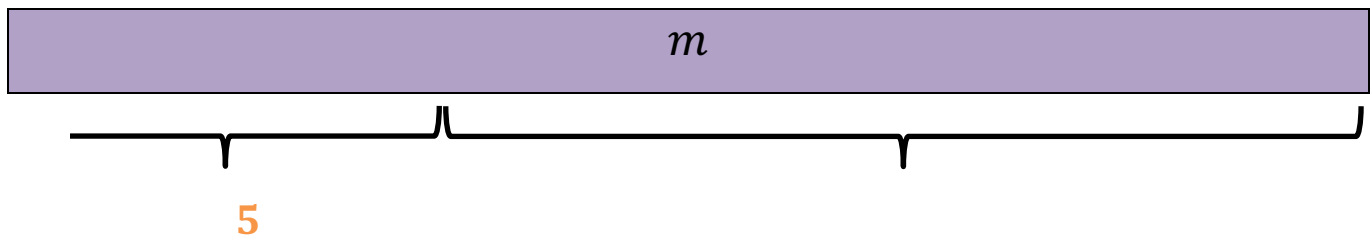
Write an algebraic expression for “The sum of a number and three.”

3. "The difference of ten and a number" can be shown where the whole bar is 10 and one part is a number represented by a variable, x . The difference is the other part.



Write an algebraic expression for "The difference of ten and a number."

4. "Five subtracted from a number" is shown with the whole bar as a number represented by a variable, m , and one part is 5. The other part is the result of 5 being subtracted from the number, m .



Complete the statement.

The algebraic expression for "Five subtracted from a number, m " is _____.

Addition/Subtraction Comparison Model

Comparison models are much like part-part-whole models. They show how much more or less one value is than another.



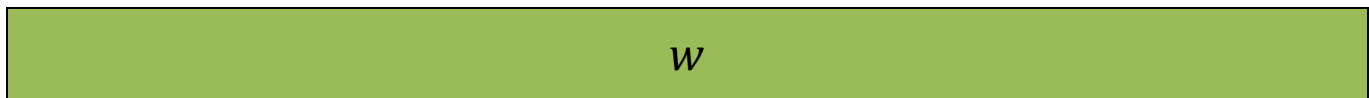
Value B

The comparison of A and B

How much **more** Value A is **than** Value B,
or

How much **less** Value B is **than** Value A

5. Write an algebraic expression for the bar model shown.



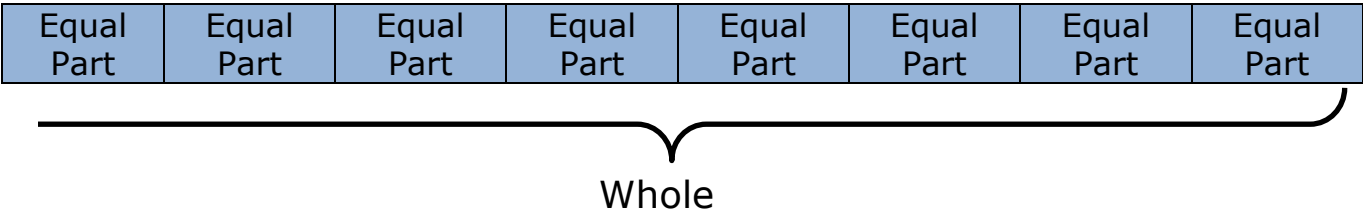
**Eight less than a
number**

8

Algebraic expression: _____

Equal Parts – Whole Model

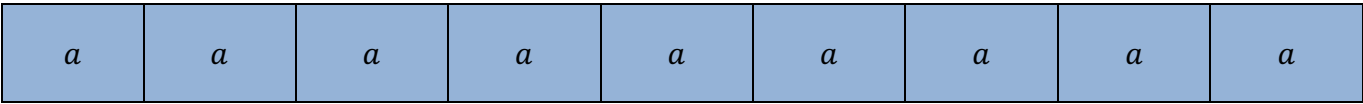
The Equal Parts – Whole Model shows the whole sectioned into equal parts.



The equal parts-whole model can be used to show the following actions.

Action	Equation
Repeated addition (multiplication)	Equal part + ... + equal part = whole Equal part × number of parts = whole
Equal sharing	Whole ÷ number of parts = equal part
Partitioning	Whole ÷ equal part = number of parts

6. “The product of nine and a number” can be shown as 9 equal parts with size of “a number” represented with the variable, *a*.

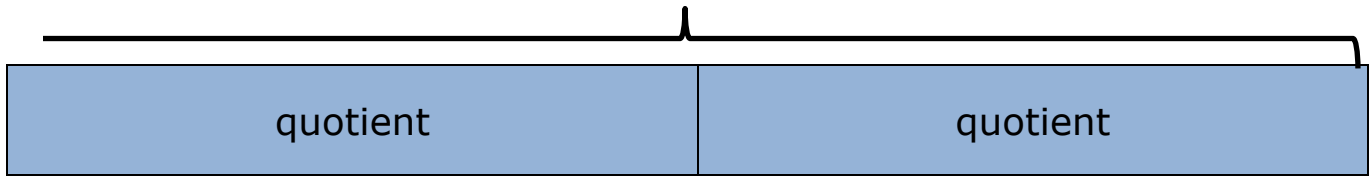


The product of nine and a number

Write an algebraic expression for “the product of nine and a number.”

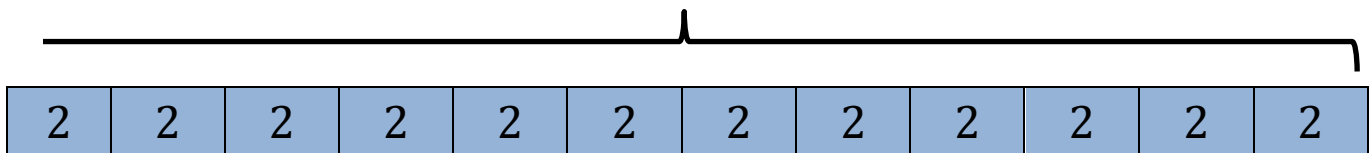
7. "The quotient of a number and two" could be shown as "a number" represented by the variable, p , sectioned off into two equal parts. The quotient is the size of each equal part.

A number, p



"The quotient of a number and two" could also be shown as "a number" represented by the variable p , sectioned off into equal parts with size of 2 units. The quotient is the number of equal parts.

A number, p



What is the algebraic expression for "The quotient of a number and 2?"

12.3.4 Collaboration: Matching Models and Writing Expressions

Part 1

Work with your partner to match each written description to the appropriate bar model, where the red portion is the result of the written description. Some models will be used more than once.

Complete the table with the letter of the appropriate bar model. Then, write the algebraic expression that matches the word description and bar model.

Written Description	Bar Model	Algebraic Expression
The difference of a number and four		
Four decreased by a number		
Four more than a number		
A number subtracted from four		
A number increased by four		
Four less than a number		
The sum of four and a number		

A.

4

n

B.

x

C.

4

w

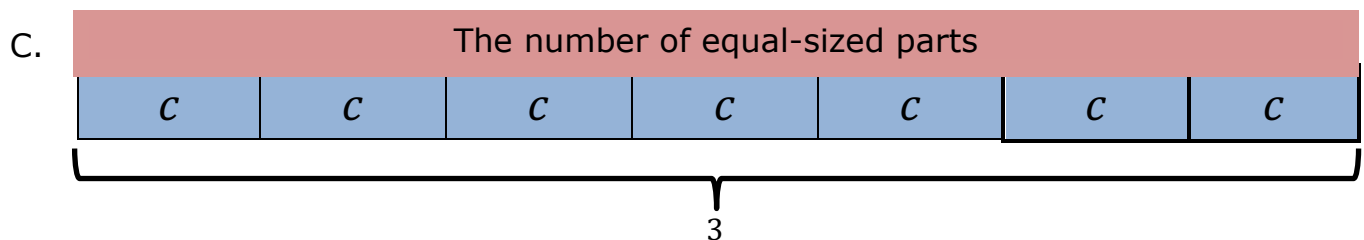
D.

p

4

Work with a different partner to match each written description to the appropriate bar model, where the red portion is the result of the written description. Some models may be used more than once.

Written Description	Bar Model	Algebraic Expression
A number tripled		
A number equally shared into three parts		
Three times a number		
The quotient of three and a number		
The product of three and a number		
A number divided by three		

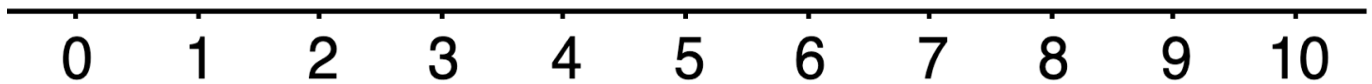


12.3.5 Guided Practice: Writing Algebraic Expressions from Written Descriptions

1. Sketch a bar model and write an algebraic expression for each written description.
 - a. The product of a number and five
 - b. Three tenths less than a number
 - c. The quotient of a number and twelve
 - d. A number doubled
 - e. The difference of fifteen and a number
2. Compare your expressions and bar models with your partner. Make any revisions if necessary.

12.3.6 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of translating a written description into an algebraic expression.



2. Complete the statement.

I chose this emoji because _____
